# Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

16 October 2026

Hiring Team
Arm

Application: Software Machine Learning Engineer (Job ID 2026-18685)

Dear Arm hiring team,

I am applying for the Software Machine Learning Engineer role on Arm's machine learning software stack. For three years at HDFC Asset Management I owned machine learning and software systems end to end - designing the architecture, building the Python services, deploying them on AWS and fixing them in production - and the mix of Python engineering, ML frameworks and performance work this role describes is what I want to keep doing. Building software that makes AI run efficiently on next-generation accelerator designs, and growing into compiler and C++ work alongside strong engineers, is where I would like to bring that experience.

At HDFC I designed and built the applications end to end: Python FastAPI microservices in Docker (uvicorn) and a ReactJS frontend, retrieval with LlamaIndex and Azure OpenAI, a separate pgvector database per department, and agents that call Python functions as tools to run SQL-backed fund comparisons and generate PDF reports on request. I shipped them on AWS through a one-click Jenkins and GitLab CI/CD pipeline, took them from repeatedly failing the security team's VAPT to passing it, and led up to four interns as a de facto team lead, which covers the mentoring and larger-work-package side of the role.

Two parts of my work map directly onto what you ask for. On performance, I profiled the fund-comparison calculation endpoints, found the Python APIs were the bottleneck and wrote ExpressJS APIs during development to serve the same calculations faster, keeping Python FastAPI for the shipping backend; and where model choice was an engineering trade-off rather than a default, I self-hosted Llama-3 on a GPU EC2 instance to control cost and latency for our largest department and kept the stronger Azure OpenAI outputs for decision makers. On machine learning, the work was hands-on even though it was LLM and retrieval rather than compiler work: retrieval and embedding pipelines, agents calling tools, and, for my MSc, VisionCaptioner, a team study comparing CNN, ResNet-18 and ViT encoders with LSTM and attention decoders across three datasets and five seeds and benchmarking them against pre-trained vision-language models.

I want to be direct about where my experience differs from your required list. My strong language is Python; I have not worked in C or C++ professionally, and I have no compiler, kernel, driver or embedded software experience. My source control is Git with GitLab and Jenkins CI/CD rather than Gerrit or Repo, and I have not worked to a formal test-driven development discipline - my testing was user feedback, Compliance review and my own checks rather than automated test suites. My deep learning is academic rather than in a production ML framework stack. What I do bring is the engineering around it: Python services and ML applications delivered end to end, performance work where it mattered, deployment and CI/CD ownership, cross-team delivery with named stakeholders, and mentoring, with about three years of experience rather than a longer track record.

I am based in Bristol and happy to discuss the Manchester or Cambridge office and the hybrid arrangement. I hold a Student Visa and am eligible for the Graduate Visa, which gives two years of full working rights, and I understand Arm covers the employer-side sponsorship costs while individual application fees are the candidate's. Thank you for considering my application; I would be glad to talk it through.

Yours sincerely,
Kaustubh Sonawane